Development of a Defect Estimation Method for CFRP Interstage Structures with Holes using Finite Element Analysis and Graph Neural Networks

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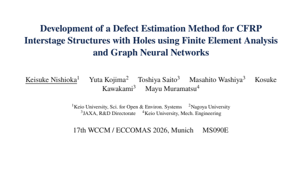
17th WCCM / ECCOMAS 2026, Munich MS090E



Good morning. I am Keisuke Nishioka, from Keio University and Leibniz Universitaet Hannover, with co-authors at Nagoya University and JAXA. Today I show how we localize hidden delamination in perforated CFRP interstage structures using finite-element analysis and a graph neural network: 61% macro-F1 from surface stress alone, and how we make it robust to measurement noise. About 15 minutes.

Outline

1. Motivation & measurement — defects in CFRP, infrared stress (DSPSS)
2. Method — difference-normalized input & a GAT on the FEM mesh
3. Results — 19-class 3-D localization (macro-F1 61%)
4. New: robustness to measurement noise (defect-free negatives)
5. Outlook

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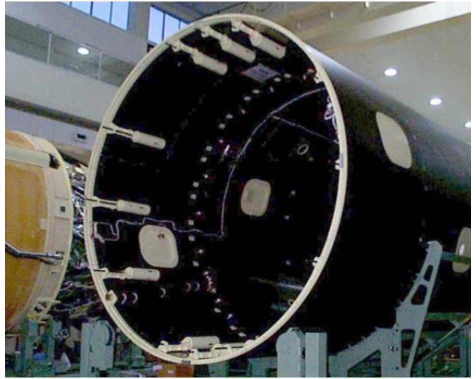
Quick roadmap: motivation and the DSPSS measurement, the method, the published results, the new noise-robustness work, and outlook. About 15 minutes.

Introduction: CFRP in space transportation

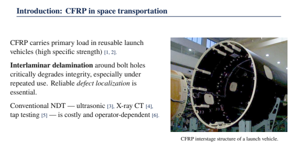
CFRP carries primary load in reusable launch vehicles (high specific strength) [1, 2].

Interlaminar delamination around bolt holes critically degrades integrity, especially under repeated use. Reliable *defect localization* is essential.

Conventional NDT — ultrasonic [3], X-ray CT [4], tap testing [5] — is costly and operator-dependent [6].



CFRP interstage structure of a launch vehicle.



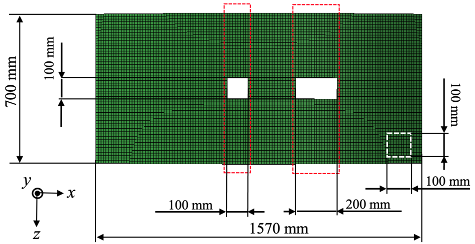
CFRP is the primary structural material in reusable launch vehicles, for its strength-to-weight ratio. Around bolt holes, interlaminar delamination accumulates and is invisible from the outside. Conventional NDT — ultrasonic, X-ray CT, tap testing — is slow, expensive, and operator-dependent. So we need an automated, reliable way to localize these sub-surface defects.

Infrared stress measurement (DSPSS)

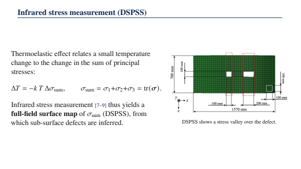
Thermoelastic effect relates a small temperature change to the change in the sum of principal stresses:

$$\Delta T = -k\,T\,\Delta\sigma_{\mathrm{sum}}, \qquad \sigma_{\mathrm{sum}} = \sigma_1+\sigma_2+\sigma_3 = \mathrm{tr}(\boldsymbol{\sigma}).$$

Infrared stress measurement [7–9] thus yields a **full-field surface map** of σ_{sum} (DSPSS), from which sub-surface defects are inferred.



DSPSS shows a stress valley over the defect.

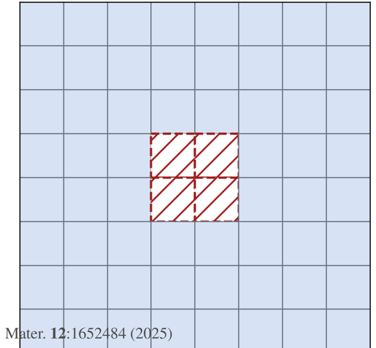


Our measurable quantity is infrared stress measurement. By the thermoelastic effect, a small temperature change is proportional to the change in the sum of principal stresses. So an infrared camera gives a full-field surface map of the stress sum, which we call DSPSS. A sub-surface defect perturbs this field and creates a local stress valley, as shown on the right.

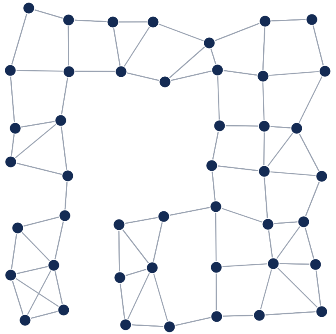
Previous study and purpose

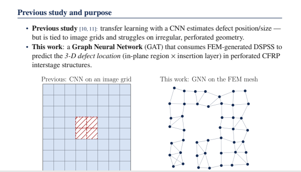
- **Previous study** [10, 11]: transfer learning with a CNN estimates defect position/size — but is tied to image grids and struggles on irregular, perforated geometry.
- **This work**: a **Graph Neural Network** (GAT) that consumes FEM-generated DSPSS to predict the *3-D defect location* (in-plane region × insertion layer) in perforated CFRP interstage structures.

Previous: CNN on an image grid



This work: GNN on the FEM mesh



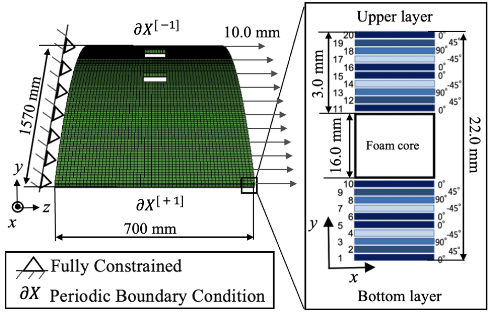


A previous study used transfer learning with a CNN to estimate defect position and size. But a CNN works on a regular image grid and struggles with the irregular, hole-perforated geometry. Our idea is to treat the FEM surface mesh directly as a graph and use a graph attention network. The task is the full 3-D location: both the in-plane region and which layer the delamination sits in.

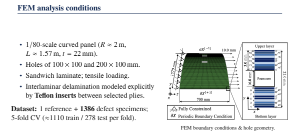
FEM analysis conditions

- 1/80-scale curved panel ($R \approx 2$ m, $L \approx 1.57$ m, $t = 22$ mm).
- Holes of 100×100 and 200×100 mm.
- Sandwich laminate; tensile loading.
- Interlaminar delamination modeled explicitly by **Teflon inserts** between selected plies.

Dataset: 1 reference + **1386** defect specimens; 5-fold CV (≈1110 train / 278 test per fold).



FEM boundary conditions & hole geometry.



We build the dataset by finite-element analysis: a one-eightieth-scale curved interstage panel, about 2 metres radius, with two rectangular holes. Delaminations are modeled explicitly by inserting thin Teflon sheets between selected plies, under tensile loading. This gives DSPSS fields with exactly known ground-truth defects.

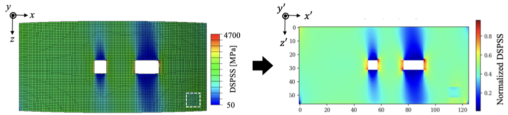
Input data: normalized / difference DSPSS

Raw DSPSS is dominated by **stress concentration around the hole**, masking the defect signal.

Remedy (paper):

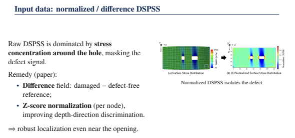
- **Difference** field: damaged − defect-free reference;
- **Z-score normalization** (per node), improving depth-direction discrimination.

⇒ robust localization even near the opening.



(a) Surface Stress Distribution (b) 2D Normalized Surface Stress Distribution

Normalized DSPSS isolates the defect.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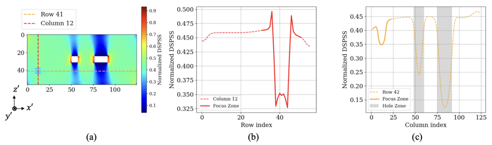
Remedy (paper):
Difference field: damaged − defect-free reference;
Z-score normalization (per node), improving depth-direction discrimination.
⇒ robust localization even near the opening.
Normalized DSPSS isolates the defect.

The raw DSPSS is dominated by the stress concentration around the holes, which masks the defect. So we subtract a defect-free reference to get a difference field, and z-score normalize per node. This isolates the defect signal and, importantly, improves discrimination in the depth direction — that is, which layer the defect is in.

Stress distribution around the defect

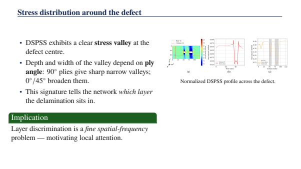
- DSPSS exhibits a clear **stress valley** at the defect centre.
- Depth and width of the valley depend on **ply angle**: 90° plies give sharp narrow valleys; 0°/45° broaden them.
- This signature tells the network *which layer* the delamination sits in.



Normalized DSPSS profile across the defect.

Implication

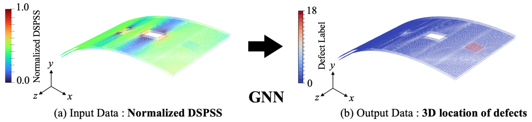
Layer discrimination is a *fine spatial-frequency* problem — motivating local attention.



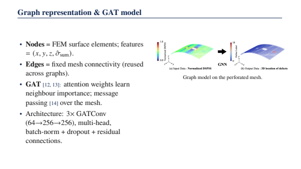
Looking closer, each defect produces a stress valley at its centre. The depth and width of that valley depend on the ply angle above it: ninety-degree plies give sharp narrow valleys, zero and forty-five broaden them. This ply-angle signature is exactly the cue the network uses to tell which layer the delamination is in. It is a fine spatial-frequency problem, which motivates local attention.

Graph representation & GAT model

- **Nodes** = FEM surface elements; features = $(x, y, z, \hat{\sigma}_{\text{sum}})$.
- **Edges** = fixed mesh connectivity (reused across graphs).
- **GAT** [12, 13]: attention weights learn neighbour importance; message passing [14] over the mesh.
- Architecture: 3× GATConv (64→256→256), multi-head, batch-norm + dropout + residual connections.



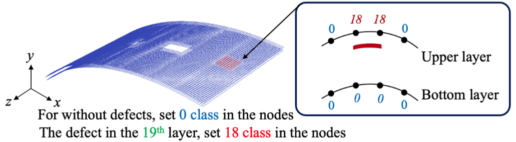
Graph model on the perforated mesh.



We represent each surface element as a node, with features being its coordinates and the normalized stress. Edges are the fixed mesh connectivity, reused across all specimens. We use a graph attention network: three GAT layers, sixty-four to two-hundred-fifty-six channels, multi-head, with batch-norm, dropout and residual connections. Attention lets each node weight its neighbours adaptively, which suits the highly non-uniform stress field.

Output: 3-D defect location as 19-class node labels

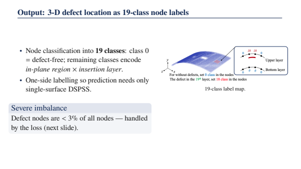
- Node classification into **19 classes**: class 0 = defect-free; remaining classes encode *in-plane region × insertion layer*.
- One-side labelling so prediction needs only single-surface DSPSS.



19-class label map.

Severe imbalance

Defect nodes are < 3% of all nodes — handled by the loss (next slide).

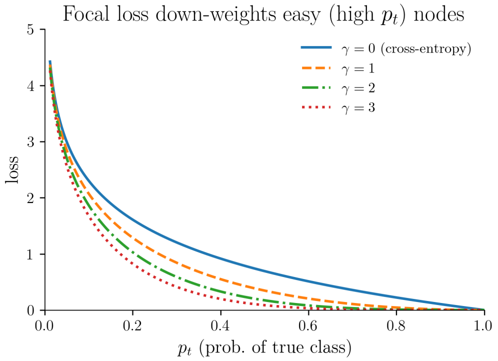


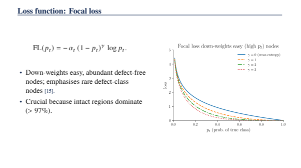
We formulate localization as node classification into nineteen classes: class zero is defect-free, and the other classes jointly encode the in-plane region and the insertion layer. We label only one side, so prediction needs a single surface measurement. Note the severe imbalance — defect nodes are under three percent of all nodes.

Loss function: Focal loss

$$\mathrm{FL}(p_t) = -\alpha_t\,(1-p_t)^{\gamma}\,\log p_t .$$

- Down-weights easy, abundant defect-free nodes; emphasises rare defect-class nodes [15].
- Crucial because intact regions dominate ($> 97\%$).



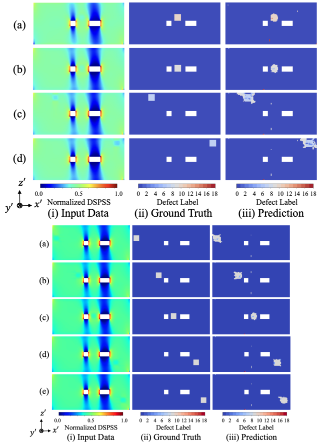


To handle that imbalance we use focal loss, which down-weights the easy, abundant defect-free nodes and focuses learning on the rare defect classes. Without it, the model simply predicts defect-free everywhere.

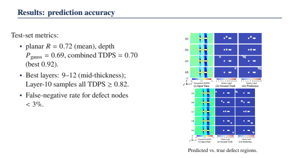
Results: prediction accuracy

Test-set metrics:

- planar $R = 0.72$ (mean), depth $P_{\text{gauss}} = 0.69$, combined TDPS = 0.70 (best 0.92).
- Best layers: 9–12 (mid-thickness); Layer-10 samples all TDPS ≥ 0.82.
- False-negative rate for defect nodes $< 3\%$.



Predicted vs. true defect regions.



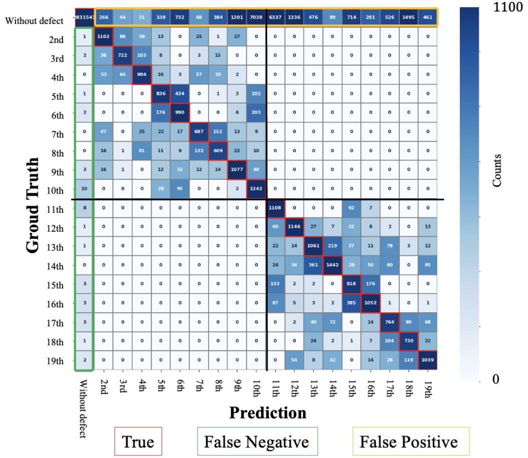
On held-out test specimens we get a planar score R of 0.72, a depth score of 0.69, and a combined TDPS of 0.70, up to 0.92 in the best case. The mid-thickness layers, nine to twelve, are localized best. Critically, the false-negative rate on defect nodes is under three percent, so we rarely miss a real defect.

Overall performance & confusion

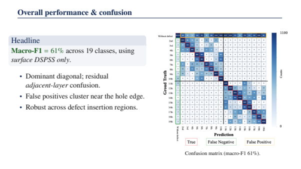
Headline

Macro-F1 = 61% across 19 classes, using *surface DSPSS only*.

- Dominant diagonal; residual *adjacent-layer* confusion.
- False positives cluster near the hole edge.
- Robust across defect insertion regions.



Confusion matrix (macro-F1 61%).



Overall macro-F1 is sixty-one percent across all nineteen classes, from surface DSPSS alone. The confusion matrix has a strong diagonal; the residual errors are mostly confusion with the adjacent layer, and false positives cluster near the hole edge, where stress gradients are steep.

Conclusion

- Introduced an **FEM + GAT** framework for 3-D delamination localization in perforated CFRP interstage panels.
- **Difference-normalized DSPSS** suppresses hole stress concentration and enables full-region estimation.
- Achieved **macro-F1** 61% on 19 classes from surface data only, with a very low defect false-negative rate and no false detections in defect-free regions.

A scalable foundation

toward experimental infrared-stress (TSA) data and practical SHM of aerospace composite structures.

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toward experimental infrared-stress (TSA) data and practical SHM of aerospace composite structures.

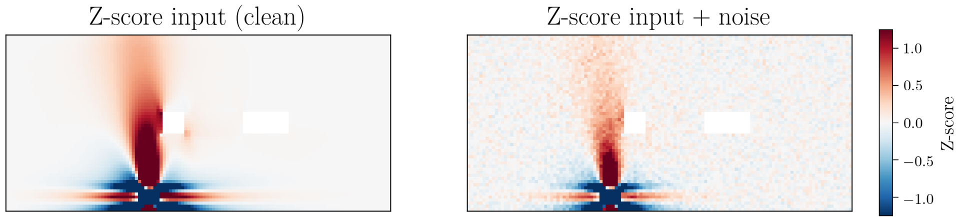
To summarize the published work: an FEM-plus-GAT framework localizes 3-D delamination in perforated CFRP; difference-normalized DSPSS removes the hole concentration and enables full-region estimation; and we reach sixty-one percent macro-F1 from surface data alone, with very low false negatives and no false detections in healthy regions.

New focus: robustness for real operation

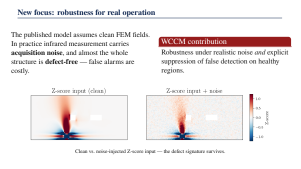
The published model assumes clean FEM fields. In practice infrared measurement carries **acquisition noise**, and almost the whole structure is **defect-free** — false alarms are costly.

WCCM contribution

Robustness under realistic noise *and* explicit suppression of false detection on healthy regions.



Clean vs. noise-injected Z-score input — the defect signature survives.

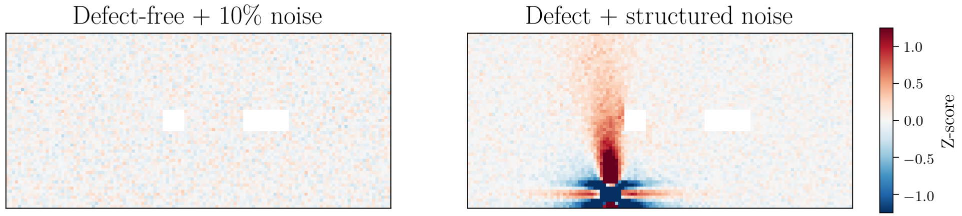


Now the new part for this congress. The published model assumes clean FEM fields. Real infrared measurements carry noise — temperature drift, emissivity, sensor noise. And in practice almost the entire structure is healthy, so false alarms are very costly. So here we focus on robustness under realistic noise and on suppressing false detection. On the right, the input with and without injected noise — the defect signature still survives.

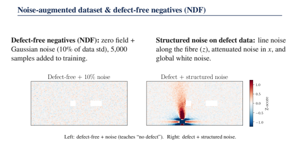
Noise-augmented dataset & defect-free negatives (NDF)

Defect-free negatives (NDF): zero field + Gaussian noise (10% of data std), 5,000 samples added to training.

Structured noise on defect data: line noise along the fibre (z), attenuated noise in x, and global white noise.



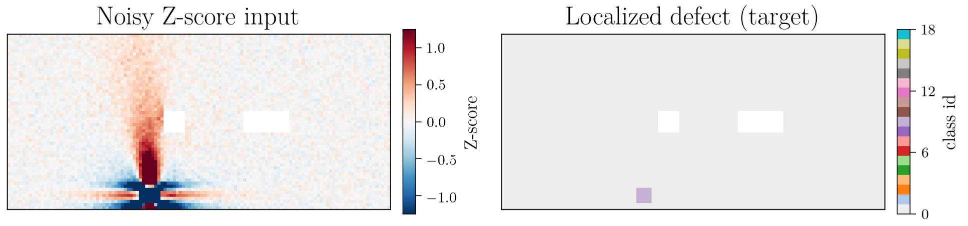
Left: defect-free + noise (teaches "no defect"). Right: defect + structured noise.



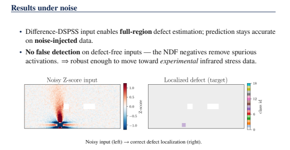
We do two things. First, defect-free negatives: we add five thousand samples that are a zero field plus Gaussian noise at ten percent of the data standard deviation, so the model explicitly learns what no-defect looks like. Second, we add structured noise to the defect data — line noise along the fibre direction, attenuated noise across it, and global white noise — to mimic measurement artefacts.

Results under noise

- Difference-DSPSS input enables **full-region** defect estimation; prediction stays accurate on **noise-injected** data.
- **No false detection** on defect-free inputs — the NDF negatives remove spurious activations. ⇒ robust enough to move toward *experimental* infrared stress data.



Noisy input (left) ⟶ correct defect localization (right).



The outcome: with the difference-DSPSS input, the model still estimates defects across the full region; prediction stays accurate on noise-injected data; and the defect-free negatives remove spurious activations, so we get essentially no false detection on healthy inputs. Here, the noisy input on the left still yields the correct target localization on the right. This makes the framework robust enough to move toward real experimental data.

Outlook [ongoing, beyond the published paper]

- **Sim-to-real:** apply the noise-robust model to *experimental* infrared-stress (TSA) measurements.
- **Architecture study (preliminary):** extending from attention GNNs to *mesh-physics* GNNs (MeshGraphNet) and layered variants; early results are encouraging and under evaluation with co-authors.
- **Application:** transfer to aerospace payload-fairing SHM.

Preliminary; not part of the published results.

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- Application: transfer to aerospace payload-fairing SHM.
Preliminary; not part of the published results.

Going forward: first, sim-to-real — applying this noise-robust model to actual infrared stress measurements. Second, and still preliminary, we are extending from attention GNNs to mesh-physics GNNs such as MeshGraphNet and some layered variants; early results are encouraging and under evaluation with our co-authors. And third, transferring the approach to payload-fairing structural health monitoring. I will stop here. Thank you — I am happy to take questions.

Acknowledgements

- Conducted in collaboration with **JAXA** (Research & Development Directorate, Research Unit IV).
- Co-authors at **Keio University**, **Nagoya University**, and **JAXA**.
- Built on the FEA-generated CFRP interstage dataset of the prior study [10].

Thank you for your attention — questions welcome.

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Thank the co-authors and JAXA, then invite questions.

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Appendix

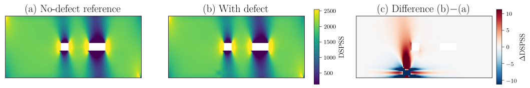
Development of a Defect Estimation Method for CFRP Interstage Structures with Holes using Finite Element Analysis and Graph Neural Networks

Appendix

Appendix: difference-data construction

Raw DSPSS near a hole is dominated by geometric stress concentration. We remove it by a per-specimen **difference**:

$$\text{Difference} = \underbrace{\text{Original (defect)}}_{\text{damaged}} - \underbrace{\text{Reference (no defect)}}_{\text{pristine}}.$$



With-defect − no-defect reference = difference field (defect localized).

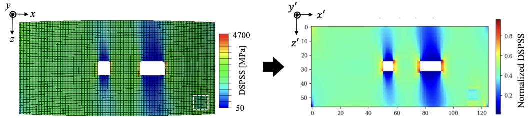
The residual isolates the defect-induced perturbation, independent of the hole's baseline field.

Appendix: Z-score normalization

The difference field is standardized per node (Z-score):

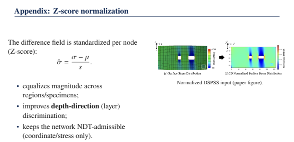
$$\hat{\sigma} = \frac{\sigma - \mu}{s}.$$

- equalizes magnitude across regions/specimens;
- improves **depth-direction** (layer) discrimination;
- keeps the network NDT-admissible (coordinate/stress only).



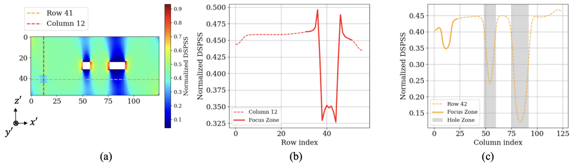
(a) Surface Stress Distribution (b) 2D Normalized Surface Stress Distribution

Normalized DSPSS input (paper figure).



Appendix: DSPSS profile around the defect

The difference-DSPSS shows a pronounced **stress valley** at the defect centre; its depth and width encode the insertion layer (ply-angle dependence).



Profile of the normalized DSPSS across the defect (paper figure).

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Profile of the normalized DSPSS across the defect (paper figure).

Appendix: evaluation metrics

- R — in-plane (planar) F1 score for defect vs. non-defect at the correct location.
- P_{gauss} — depth-direction score: Gaussian-weighted ($\sigma = 2$ layers) credit for predicting the correct or a near-correct insertion layer.
- **TDPS** (Total Defect Prediction Score) = $\frac{1}{2}\left(R + P_{\text{gauss}}\right)$.
- **macro-F1** — unweighted mean F1 over all 19 classes (the headline metric).

These quantify in-plane localization (R), depth/layer accuracy (P_{gauss}), and their combination (TDPS), complementing the class-level macro-F1.

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